

Cloud Computing

Assignment # 01

LAB TITLE: Git, Gitea, GitHub, Git LFS, and GitHub Pages

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Task 1 – Gitea Setup and Repository

1.1 Gitea Installation and Running Containers

```
sananoorbhatti@ubuntu:~$ cd Gitea
sananoorbhatti@ubuntu:~/Gitea$ ls
compose.yaml
sananoorbhatti@ubuntu:~/Gitea$ docker compose up -d
[+] up 25/25
V Image gitea/gitea:latest Pulled 50.5s
V Image postgres:17-alpine Pulled 55.9s
V Network webnet Created 0.2s
V Volume gitea_postgres Created 0.0s
V Volume gitea Created 0.0s
V Container gitea Started 2.3s
V Container gitea_db Started 1.4s
sananoorbhatti@ubuntu:~/Gitea$ docker compose ps
NAME          IMAGE          COMMAND                  SERVICE    CREATED      STATUS      PORTS
gitea         gitea/gitea:latest "/usr/bin/entrypoint..." gitea      2 minutes ago Up 2 minutes 22/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
gitea_db      postgres:17-alpine "docker-entrypoint.s..." db         2 minutes ago Up 2 minutes 5432/tcp
sananoorbhatti@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
sananoorbhatti@ubuntu:~/Gitea$ _
```

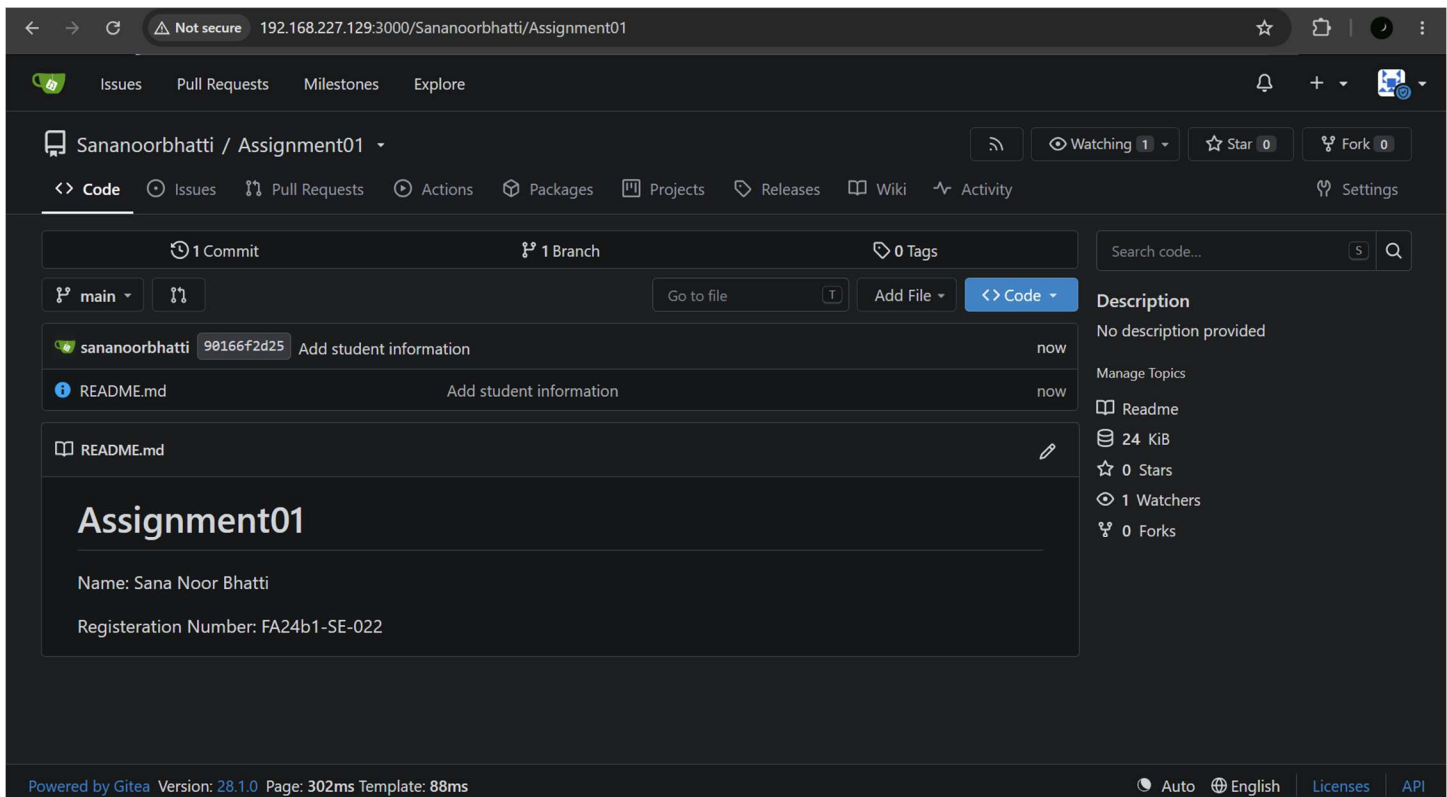
Gitea and PostgreSQL were verified as running through Docker Compose, and the HTTP response confirmed that Gitea was accessible.

1.2 Pushing the Local Repository to Gitea

```
sananoorbhatti@ubuntu:~/Assignment01$ git remote add gitea http://sananoorbhatti@127.0.0.1:3000/sananoorbhatti/Assignment01.git
sananoorbhatti@ubuntu:~/Assignment01$ git remote -v
gitea      http://sananoorbhatti@127.0.0.1:3000/sananoorbhatti/Assignment01.git (fetch)
gitea      http://sananoorbhatti@127.0.0.1:3000/sananoorbhatti/Assignment01.git (push)
sananoorbhatti@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://sananoorbhatti@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 296 bytes | 98.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/sananoorbhatti/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
sananoorbhatti@ubuntu:~/Assignment01$ _
```

The local Git repository was connected to Gitea using a remote, and the main branch was pushed successfully.

1.3 Gitea Repository Verification



The public Assignment01 repository was verified on Gitea, showing the project README and repository contents.

Task 2 – GitHub Repository and Remotes

2.1 Configuring Gitea and GitHub Remotes

```
sananoorbhatti@ubuntu:~/Assignment01$ cd ~/Assignment01
sananoorbhatti@ubuntu:~/Assignment01$ git remote add github https://github.com/sananoorbhatti/Assignment01.git
sananoorbhatti@ubuntu:~/Assignment01$ git remote -v
gitea      http://sananoorbhatti@127.0.0.1:3000/sananoorbhatti/Assignment01.git (fetch)
gitea      http://sananoorbhatti@127.0.0.1:3000/sananoorbhatti/Assignment01.git (push)
github     https://github.com/sananoorbhatti/Assignment01.git (fetch)
github     https://github.com/sananoorbhatti/Assignment01.git (push)
sananoorbhatti@ubuntu:~/Assignment01$ _
```

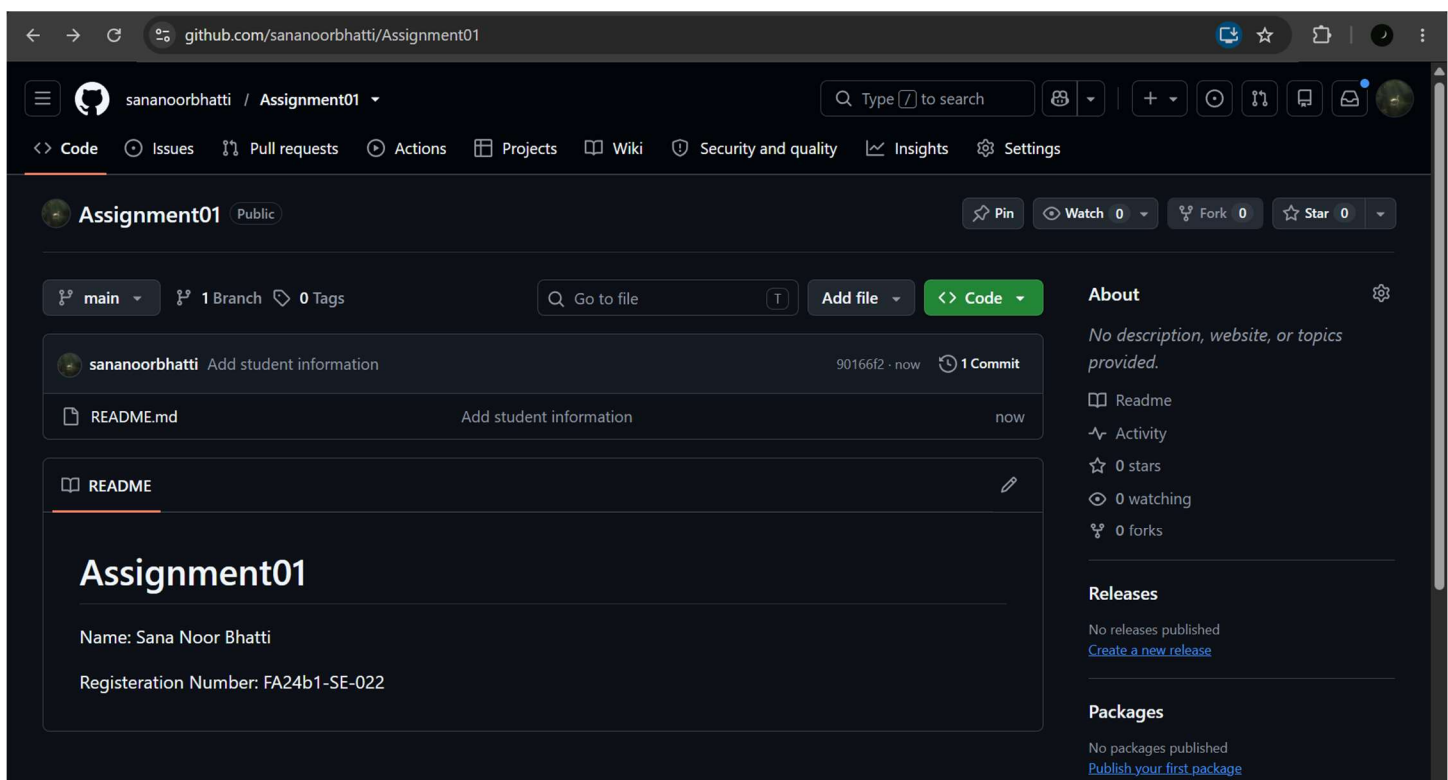
Separate remotes named gitea and github were configured and verified using the **git remote -v** command.

2.2 Pushing the Repository to GitHub

```
sananoorbhatti@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': sananoorbhatti
Password for 'https://sananoorbhatti@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 296 bytes | 296.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sananoorbhatti/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
sananoorbhatti@ubuntu:~/Assignment01$
```

The local repository was pushed to the public GitHub Assignment01 repository, successfully uploading the main branch.

2.3 GitHub Repository Verification



The screenshot shows the GitHub repository page for 'sananoorbhatti / Assignment01'. The repository is public and has 1 branch (main) and 0 tags. The main branch is selected, showing a commit by 'sananoorbhatti' with the message 'Add student information' and a commit hash of '90166f2'. The commit is dated 'now' and has 1 commit. The repository contains a file named 'README.md' with the same commit message and date. The README content is displayed below the file list, showing the title 'Assignment01' and the following text: 'Name: Sana Noor Bhatti' and 'Registration Number: FA24b1-SE-022'. The right sidebar shows the repository's statistics: 0 stars, 0 watching, and 0 forks. It also includes sections for 'About' (No description, website, or topics provided), 'Releases' (No releases published), and 'Packages' (No packages published).

The GitHub repository was opened to verify that the README and project contents were uploaded correctly.

Task 3 – Git Large File Storage (Git LFS)

3.1 Installing and Configuring Git LFS

```
sananoorbhatti@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
sananoorbhatti@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
sananoorbhatti@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
sananoorbhatti@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
sananoorbhatti@ubuntu:~/Assignment01$
```

Git LFS was initialized, and the .gitattributes file was configured to track files with the .bin extension.

3.2 Tracking Large Binary Files

```
sananoorbhatti@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
sananoorbhatti@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
sananoorbhatti@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
sananoorbhatti@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
sananoorbhatti@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
sananoorbhatti@ubuntu:~/Assignment01$ _
```

Three large binary files were created, staged, and verified using `git lfs ls-files --size` to confirm that Git LFS was tracking them.

3.3 Pushing Git LFS Files to GitHub

```

sananoorbhatti@ubuntu:~/Assignment01$ git commit -m "Add three files using Git LFS"
[main 53bc6ee] Add three files using Git LFS
4 files changed, 10 insertions(+)
create mode 100644 .gitattributes
create mode 100644 large-file-1.bin
create mode 100644 large-file-2.bin
create mode 100644 large-file-3.bin
sananoorbhatti@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': sananoorbhatti
Password for 'https://sananoorbhatti@github.com':
Username for 'https://github.com': sananoorbhatti
Password for 'https://sananoorbhatti@github.com':
Enumerating objects: 7, done.3/3), 321 MB | 4.6 MB/s
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 793 bytes | 793.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sananoorbhatti/Assignment01.git
90166f2..53bc6ee main -> main
sananoorbhatti@ubuntu:~/Assignment01$

```

The large files were committed and pushed to GitHub, and the Git LFS upload completed successfully.

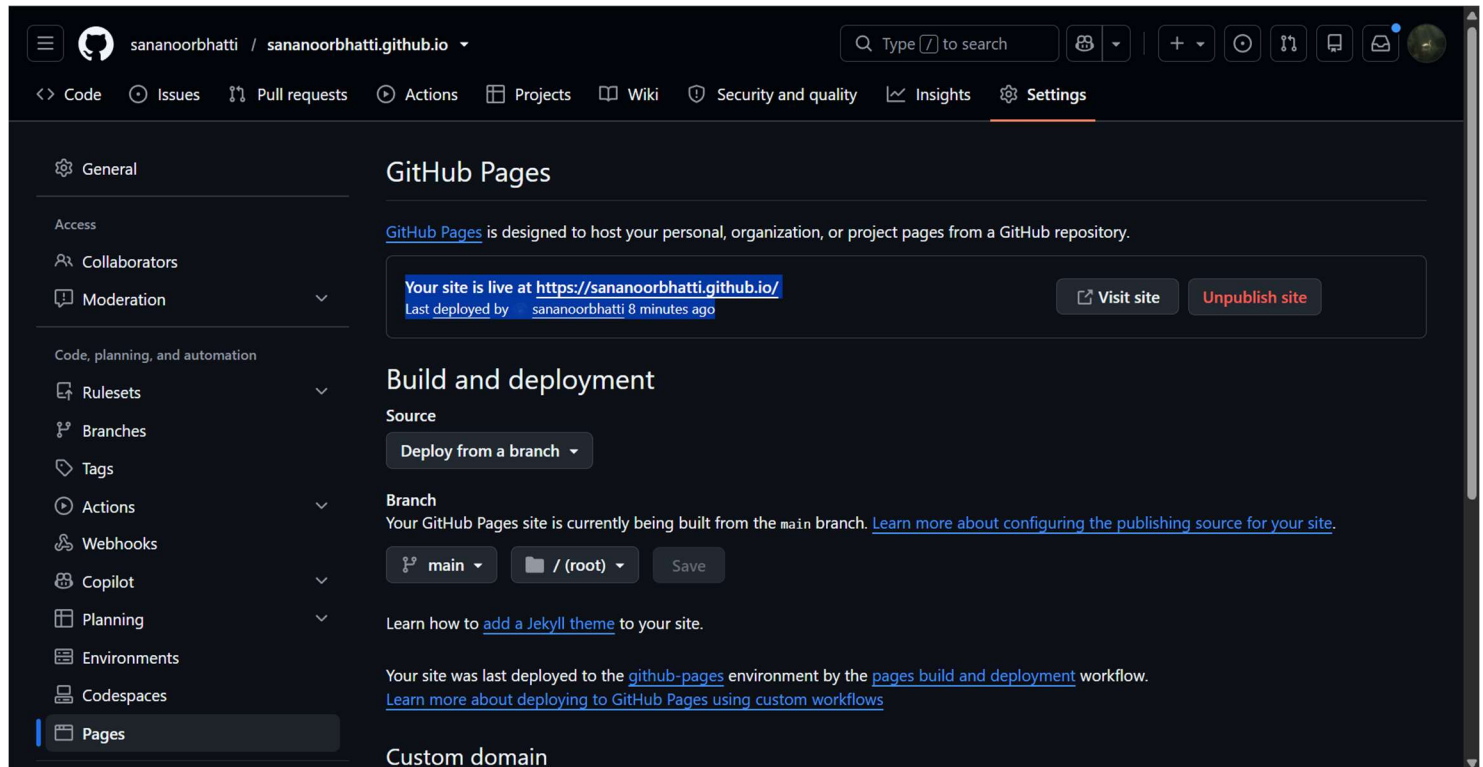
Task 4 – GitHub Pages Portfolio

4.1 Portfolio Repository and Website Files

The screenshot shows the GitHub repository page for `sananoorbhatti.github.io`. The repository is public and has 1 branch (main) and 0 tags. The commit history shows a recent commit `Update styles.css` by `sananoorbhatti` with a commit hash of `d014a81` 7 minutes ago. The commit message is `Update styles.css`. The commit history also shows previous commits for `README.md` and `index.html`. The README file is visible and contains the text `sananoorbhatti.github.io`. The right sidebar shows the repository's statistics: 0 stars, 0 forks, and 0 watchers. The 'Releases' section shows no releases published. The 'Deployments' section shows 22 deployments, with the latest one being `github-pages` 7 minutes ago.

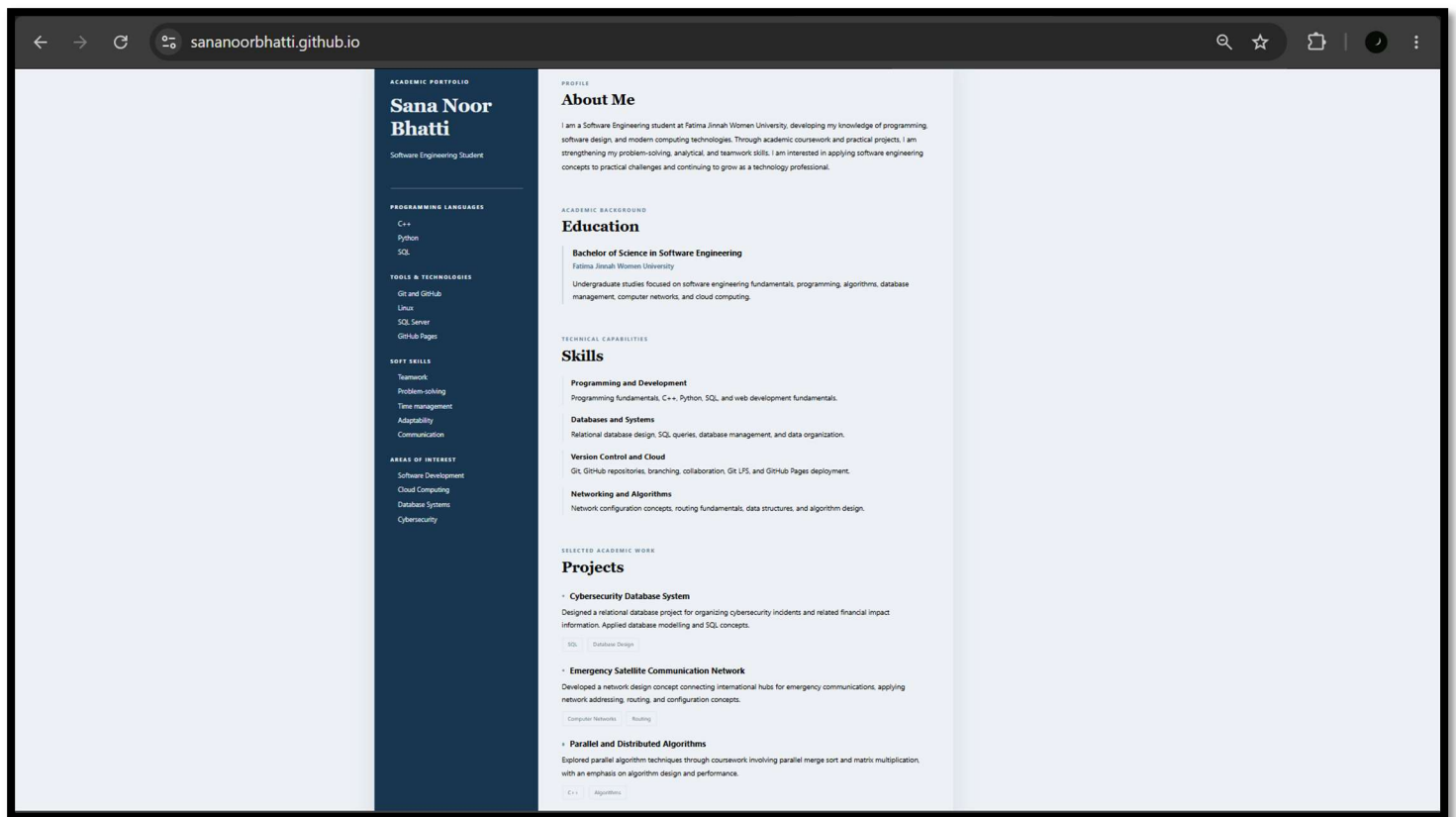
The **sananoorbhatti.github.io** repository was configured with the portfolio's `index.html` and `styles.css` files.

4.2 GitHub Pages Deployment Configuration



GitHub Pages was configured to deploy the portfolio from the main branch using the / (root) directory. The deployment status confirmed that the website was successfully published.

4.3 Live Portfolio Website



The portfolio website was opened successfully at **<https://sananoorbhatti.github.io/>**, displaying the About Me, Education, Skills, and Projects sections.

~ Assignment 01 Executed